

Representation theory of p -adic groups and the Bernstein centre

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Lecture 1

$F/\mathbb{Q}_p$ a finite ext'n. G/F reductive group, coeffs of rep = $\mathbb{C}$

Def. A rep (π, V) of $G(F)$ is smooth if $\forall v \in V, \exists \text{ c.o. } K \subset G(F) \text{ s.t. } v \in V^K$

(π, V) is admissible if $\dim V^K \stackrel{\text{smooth}}{+} < \infty, \forall \text{ c.o. } K$.

$\text{Rep}(G)$ cat. of smooth reps of $G(F)$.

Extended example In smooth reps of $GL_2(\mathbb{Q}_p)$

- characters e.g. $|\cdot|_p \circ \det$

$$(GL_2^x)^2 = T(\mathbb{Q}_p) \leftarrow \underset{\text{upper } \Delta}{B(\mathbb{Q}_p)} \rightarrow G(\mathbb{Q}_p)$$

$$x_1 \boxtimes x_2 \rightsquigarrow \text{Ind}_B^{GL_2}(x_1 \boxtimes x_2)$$

$$\text{Ind}_B^{GL_2} p := \left\{ f: G(\mathbb{Q}_p) \rightarrow V : f(bg) = p(b)f(g) \right\}_{sm}$$

(p, V)

$$(g \cdot b)(x) = f(xg)$$

- $\text{Ind}_B^{GL_2} p$ irr. unless $x_1 x_2^{-1} = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}$

- If $x_1 = x_2 = \mathbb{1}$, $\mathbb{1} \rightarrow \text{Ind}_B^G \mathbb{1} \rightarrow \underline{\text{st}}$
irr.

- Supercuspidals

Defn $\pi \in \text{Rep}(G)$ is s.c. if $\text{Hom}(\pi, \text{Ind}_P^G p) = 0, \forall \text{ proper } P, p$.

Expect (known in many cases) $\text{irr. sic. } \pi \simeq \text{c-ind}_{\tilde{K}}^G P$, $\tilde{K}$ compact mod centre
 P irr. rep of $\tilde{K}$

$$\downarrow$$

$$\left\{ f \in \text{Ind}_{\tilde{K}}^G P : f \text{ compact supp. mod } \tilde{K} \right\}$$

Ex. $\text{c-ind}_{G_2(\mathbb{Z}_p)}^{G_2(\mathbb{Q}_p)} \tilde{\rho}$

s.t. $\tilde{\rho}$ an ext'n of a cusp. rep of $G_2(\mathbb{F}_p)$ lifted to $G_2(\mathbb{Z}_p)$.

More properties of induction

- Renormalize $I_P^G := \text{Ind}_P^G \cdot (\otimes S_P^{1/2})$

1) I_P^G , $\text{c-ind}_{\tilde{K}}^G$ exact

2) $I_P^G : \text{adm.} \mapsto \text{adm.}$

3) I_P^G has a left adjoint $J_P^G := \delta_P^{-1/2} \text{Jac}_P^G$

$$\text{Jac}_P^G : \pi \mapsto (\pi|_P)_U, \quad P = MU$$

4) I_P^G has a right adjoint $J_{P^{\text{op}}}^G$.

5) c-ind has right adjoint res.

Def. The contragredient $(\pi, V)^\vee$ of (π, V) is $V_{S_m}^*$.

- π adm. $\Leftrightarrow (\pi^\vee)^\vee \simeq \pi$

- $I_P^G(\pi^\vee) \simeq (I_P^G \pi)^\vee$



Hecke Algebras Define $\mathcal{H}(G)$ alg of locally constant compactly supp dist on $G(F)$

After fixing a Haar measure, identify with loc. const. c-supp. functions $f \mapsto f \delta x$

Then $\mathcal{H}(G) = \bigcup_K \mathcal{H}(G, K) \leftarrow K\text{-bi-inv.}$
 $\text{End}_G \left(\text{c-Ind}_K^G \mathbb{1} \right)$

(π, V) is smooth, then $\mathcal{H}(G) \curvearrowright V$ by $\pi(f) \cdot v = \int_G f(g) \pi(g) v dg$

Th. Action induces an equiv.

$$\begin{array}{ccc} \text{Rep}(G) & \longleftrightarrow & \mathcal{H}(G)\text{-Mod}^{nd} \\ & & \left[\begin{array}{c} K \cdot M = M \\ \text{nd} \end{array} \right] \end{array}$$

$\xrightarrow{\quad}$

Cohomological duality (Bernstein)

Denote $D(\text{Rep}(G))^\omega$ for compact objects in $D(\text{Rep}(G))$

Bernstein: A^* compact $\Leftrightarrow$ $H^i(A^*)$ f.g.
 $-$ odd coh. amplitude

Define. $D_{\text{coh}} : \pi \mapsto \text{RHom}_G(\pi, \mathcal{H}(G))$ a duality on $D(\text{Rep}(G))^\omega$

Th

- π irred., $\Rightarrow D_{\text{coh}}(\pi)$ is supp. in a single degree $d\pi$
- $D_{\text{coh}}(\pi)[+d\pi]$ is irred.
- $D_{\text{coh}}(\pi)[d\pi] \simeq \text{AZ}(\pi)^\vee$

AZ: Aubert - Zelevinsky duality

$$\text{AZ}(\pi) := \pm \sum_{M \text{ std}} (-1)^{\# \Delta_M} I_P^G J_P^G \pi$$

Ex. - π s.c. $\int_P^G \pi = 0$ when P proper, $AZ(\pi) = \pi$

- $G = GL_2$, $\pi = St$

St $P = G$

$-I_P^G \chi$ $P = B$

($1 + St$ in Gr_{oth})

$$D_{coh}(St) = \mathbb{1}[-2]$$

Lecture 2. Def. The Bernstein centre is $Z_{Rep(G)} := \text{End}_{Rep(G)}(id)$.

By Schur's Lemma, π irr. $z \in Z_{Rep(G)}$, then $z \propto \pi$ by a scalar.

$$Z_{Rep(G)} \simeq \varprojlim_K Z(K(G, K))$$

2 special cases

- $G(F)$ compact $\Rightarrow$ profinite $Z_{Rep(G)} = \overline{\prod_{\pi \text{ irr.}}} \mathbb{C}$

- $Rep(\mathbb{Z})$, $G_p^X \simeq \mathbb{Z} \times \text{compact}$.
 $s \in \mathbb{C}^X$

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\chi^{univ}} & \mathbb{C}[\mathbb{Z}] \\ & \searrow \chi_s & \downarrow s \\ & & \mathbb{C} \end{array}$$

$z \in Z_{Rep(\mathbb{Z})}$, $z \propto \chi^{univ}$ acts by elt of $\mathbb{C}[\mathbb{Z}]$.

$$Z_{Rep(\mathbb{Z})} = \mathbb{C}[\mathbb{Z}] \simeq \mathcal{O}_{\mathbb{C}^X}$$

General structure. Def. $G^o(F)$, $\pi: G \rightarrow \overbrace{G/G_{der}}^{\text{torus}}$
 $= \pi^{-1}(\underbrace{G/G_{der}(F)^o}_{\text{max'l compact}})$

A char χ of $G(F)$ is unramified if it vanishes on $G^o(F)$.

Span of unram. chars is an alg. $\mathbb{C}$ -tors $X^{unr}(G)$.

A cuspidal datum (M, σ) M a Levi, σ irr. s.c. of $M(F)$

An inertial equiv. class $[M, \sigma]$: $(M, \sigma) \sim (M', \sigma')$ if
 $M \sim M'$ conj. induces $\sigma \simeq \sigma' \chi$, χ unram M

$$\text{Th (Bernstein)} \quad Z_{\text{Rep}(G)} \simeq \prod_{[M, \sigma]} \mathcal{O}_{T[M, \sigma]}^{a_{\text{fin}}}$$

$$T_{[M, \sigma]} = X^{unr}(M) / \text{Stab}(\sigma) W$$

$$\text{Stab}(\sigma) \quad X^{unr}(M) \xrightarrow{\text{res}} X^{unr}(Z(M)^o)$$

$\cap$
ker res

$$GL_2 \quad \theta: 1 \in \mathbb{Z} \mapsto -1$$

Consequence. $\text{Rep}(G) \simeq \prod_{[M, \sigma]} \underbrace{\text{Rep}(G)_{[M, \sigma]}}_{\text{Bernstein blocks}}$

Example. $\text{Rep}(G)_{[T, 1]}$: principal block (G split)

Th $I_P^G(\sigma \chi^{univ})$ is a projective generator for $\text{Rep}(G)_{[M, \sigma]}$

Pt prj: first show $\sigma \chi^{univ}$ prj. as a rep of M , then $I_P^G: \text{prj.} \mapsto \text{prj.}$

(2nd adj, I_P^G exact)

generator: π irr. $\in \text{Rep}(G)_{[M, \sigma]}$, $I_P^G(\sigma \chi^{univ}) \xrightarrow{\text{Hom}(\pi, \cdot)} \pi$

$$\begin{array}{c} \mathbb{I}_p^G \quad \mathbb{J}_{p \times p}^G \quad \pi \longrightarrow \pi \\ \underbrace{\hspace{1.5cm}} \\ \subset \rho \text{ invd.} \end{array}$$

$$\mathbb{I}_p^G \rho \longrightarrow \pi$$

$$\sigma \chi^{\text{univ}} \longrightarrow \rho$$

$$\text{Hence, } \text{Rep}(G)_{[M, \sigma]} \simeq \text{End}_G(\mathbb{I}_p^G \sigma \chi^{\text{univ}}) - \text{mod}$$

$$\underline{\text{Claim.}} \quad (\mu = T, \sigma = \mathbb{1}) \quad \mathbb{I}_B^G \chi^{\text{univ}} \simeq \text{c-ind}_I^G \mathbb{1}$$

$$\begin{aligned} \Rightarrow \text{Rep}(G)_{[T, \mathbb{1}]} &\simeq \text{End}_G(\text{c-ind}_I^G \mathbb{1}) - \text{mod} \\ &\simeq \mathcal{H}(G, I) - \text{mod} \end{aligned}$$

A bit of type theory.

Def. A pair (K, τ) K a c.o. subgrp of $G(F)$, τ invd rep of K

is a $[M, \sigma]$ -type if $\exists \pi \in \text{Rep}(G)_{[M, \sigma]} \Leftrightarrow \text{Hom}_K(\tau, \pi) \neq 0$.

Fact. If $[G, \sigma]$ a block, $\sigma = \text{c-ind}_{\tilde{K}}^G \tau$, $\tilde{K}$ compact mod centre

Then $(\tilde{K}, \tau)$ is $[G, \sigma]$ type.

$$\begin{aligned} \text{Rep}(G)_{[M, \sigma]} &\simeq \mathcal{H}(G, \tilde{K}, \tau) - \text{mod} \\ &\simeq \text{End}_G(\text{c-ind}_{\tilde{K}}^G \tau) \end{aligned}$$

G-covers

Def (Bushnell - Kutzko) $M \subset G$ Levi (K, τ)
s.t. $K_M = K \cap M$ (K_M, τ_M)

- (K, τ) is a G-cover of (K_M, τ_M) if $\tau|_{K_M} \simeq \tau_M$,
 $\forall p$ parabolic containing M , $p = MU$, $K = (\underbrace{K \cap U^{op}}_{\text{parab}}) K_M (K \cap U)$

$\tau_{K \cap U^{\text{op}}}, \tau_{K \cap U}$ are trivial + technical condition.

$(I, 1)$ a G -coset of $(T^0, 1)$

Th (Bushnell-Kutzko) $[K_M, \tau_M]$ a $[M, \sigma]$ M -type

(K, τ) a G -coset of $(K_M, \tau_M) \Rightarrow [K, \tau]$ a $[M, \sigma]$ G -type

$$\begin{array}{ccc} \text{Rep}(G)_{[M, \sigma]} & \xrightarrow[\sim]{(\)_{\tau}} & \mathcal{H}(G, K, \tau)\text{-mod} \\ \uparrow & \curvearrowright & \uparrow - \otimes_{\mathcal{H}(M, K_M, \tau_M)} \mathcal{H}(G, K, \tau) \\ \text{Rep}(M)_{[M, \sigma]} & \xrightarrow[\sim]{(\)_{\tau_M}} & \mathcal{H}(M, K_M, \tau_M)\text{-mod} \end{array}$$

$$\underline{\text{Ex.}} \quad I_{\text{pop}}^G \left(c\text{-ind}_{K_M}^M \tau_M \right)_{\tau} \cong c\text{-ind}_{K_M}^M (\tau_M)_{\tau_M} \otimes_{\mathcal{H}(M, K_M, \tau_M)} \mathcal{H}(G, K, \tau)$$

$$\cong \mathcal{H}(M, \tau_M) \otimes_{\mathcal{H}(M, \tau_M)} \mathcal{H}(G, \tau)$$

$$\cong \mathcal{H}(G, \tau)$$

$$* \quad I_{\text{pop}}^G \left(c\text{-ind}_{K_M}^M \tau_M \right) \cong c\text{-ind}_K^G \tau$$

$$I_B^G \left(\underbrace{c\text{-ind}_{T^0}^T 1}_{\chi^{\text{univ}}} \right) \cong c\text{-ind}_I^G 1$$

End